MATH-1014 T10B
Midterm Exercise (2)

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March 16, 2026

Polar Curves: Arc Length

Problem 11

A curve in the xy-plane is defined by the polar equation $r = \sqrt{\sec 2\theta}$, where $0 \leq \theta \leq \frac{\pi}{6}$. which of the following definite integrals is the arc length of the curve?

Hint for Solution:

Recall the arc length formula for a curve given in polar coordinates $r = f(\theta)$ from $\theta = a$ to $\theta = b$:

$$L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

You will need to compute $\frac{dr}{d\theta}$ using the chain rule and substitute it into the formula.

Solution to Problem 11

Arc Length Formula: $L = \int_a^b \sqrt{r^2 + (r')^2}\, d\theta$

1. **Derivative (r'):**

$$r = (\sec 2\theta)^{\frac{1}{2}} \implies r' = \frac{1}{2}(\sec 2\theta)^{-\frac{1}{2}}(2\sec 2\theta \tan 2\theta) = \sqrt{\sec 2\theta}\tan 2\theta$$

2. **Integrand Simplification:**

$$\begin{aligned} r^2 + (r')^2 &= \sec 2\theta + \sec 2\theta \tan^2 2\theta \\ &= \sec 2\theta(1 + \tan^2 2\theta) = \sec^3 2\theta \end{aligned}$$

3. **Result:**

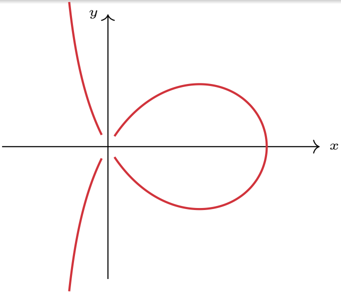
$$\boxed{L = \int_0^{\frac{\pi}{6}} \sqrt{\sec^3 2\theta}\, d\theta = \int_0^{\frac{\pi}{6}} (\sec 2\theta)^{\frac{3}{2}}\, d\theta}$$

Polar Curves: Area

Problem 14 (a) & (b)

[15 pts] Consider the curve defined by the equation $y^2 = x^2\left(\frac{3-x}{1+x}\right)$.

(a) Show that the polar equation of the curve can be written as
$r = (3\cos^2\theta - \sin^2\theta)\sec\theta$. [4 pts]
(b) Find the area inside the loop of the given curve. [7 pts]
(Hint: At what θ will the curve hit the origin?)

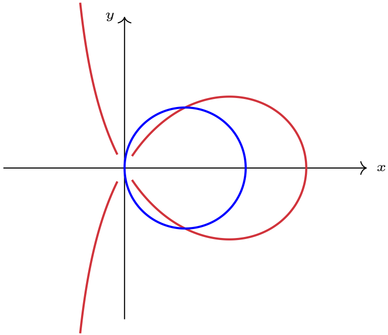


Polar Curves: Area between curves

Problem 14 (c)

(c) Use integral(s) to express the area inside the loop of the given curve but outside the circle given by the polar equation $r = 2\cos\theta$. **Do not evaluate the integral.** $[4$ pts$]$

(Hint: At what θ will the curves intersect each other?)



Solution to 14 (a) & (b)

(a) Polar Form: Let $x = r\cos\theta, y = r\sin\theta$.

$$r^2\sin^2\theta(1 + r\cos\theta) = r^2\cos^2\theta(3 - r\cos\theta)$$
$$r\cos\theta(\sin^2\theta + \cos^2\theta) = 3\cos^2\theta - \sin^2\theta$$
$$r\cos\theta = 3\cos^2\theta - \sin^2\theta \implies \boxed{r = (3\cos^2\theta - \sin^2\theta)\sec\theta}$$

Alternative simplified form: $r = 4\cos\theta - \sec\theta$.

(b) Loop Area: Hits origin when $r = 0 \implies 4\cos^2\theta = 1 \implies \theta = \pm\frac{\pi}{3}$.

$$\begin{aligned} \text{Area} &= 2 \times \frac{1}{2}\int_0^{\frac{\pi}{3}} (4\cos\theta - \sec\theta)^2\, d\theta \\ &= \int_0^{\frac{\pi}{3}} (8\cos 2\theta + \sec^2\theta)\, d\theta \quad \text{(after simplifying)} \\ &= [4\sin 2\theta + \tan\theta]_0^{\frac{\pi}{3}} = \boxed{3\sqrt{3}} \end{aligned}$$

Solution to 14 (c)

1. Intersection Points:

Set $r_{curve} = r_{circle}$:

$$4\cos\theta - \sec\theta = 2\cos\theta$$

$$2\cos\theta = \sec\theta \implies 2\cos^2\theta = 1 \implies \theta = \pm\frac{\pi}{4}$$

2. Area Integral:

Using symmetry across the polar axis:

$$\text{Area} = 2 \times \frac{1}{2}\int_0^{\frac{\pi}{4}} \left(r_{curve}^2 - r_{circle}^2\right) d\theta$$

$$\text{Area} = \boxed{\int_0^{\frac{\pi}{4}} \left((4\cos\theta - \sec\theta)^2 - (2\cos\theta)^2\right) d\theta}$$

Reduction Formulas (1)

Part III

Problem 1 (a) & (b)

([25 points]) Let $I_n = \int_0^2 \frac{1}{(x^2+4)^n}\, dx$, where $n = 1, 2, 3, \ldots$ is a positive integer.

(a) Using integration by parts, or otherwise, find $A(n), B(n)$, which are expressions depending on n, such that

$$I_{n+1} = A(n) I_n + B(n).$$

(Hint: Start with I_n.) [12 pts]

(b) Using (a), or otherwise, evaluate the integral I_2 [7 pts]

Solution to Problem 1 (a) & (b)

(a) Reduction Formula: Let $u = \frac{1}{(x^2+4)^n}$ and $dv = dx$.
Then $du = \frac{-2nx}{(x^2+4)^{n+1}}\,dx$ and $v = x$. Apply IBP on I_n:

$$I_n = \left[\frac{x}{(x^2+4)^n}\right]_0^2 - \int_0^2 \frac{-2nx^2}{(x^2+4)^{n+1}}\,dx$$

$$I_n = \frac{2}{8^n} + 2n\int_0^2 \frac{x^2+4-4}{(x^2+4)^{n+1}}\,dx$$

$$I_n = \frac{2}{8^n} + 2n\Big(I_n - 4I_{n+1}\Big) = \frac{2}{8^n} + 2nI_n - 8nI_{n+1}$$

Thus, $A(n) = \frac{2n-1}{8n}$ and $B(n) = \frac{1}{4n\cdot 8^n}$.

(b) Evaluate integral (find I_2): First, compute I_1:

$$I_1 = \int_0^2 \frac{1}{x^2+4}\,dx = \left[\frac{1}{2}\arctan\left(\frac{x}{2}\right)\right]_0^2 = \frac{\pi}{8}$$

$$I_2 = A(1)I_1 + B(1) = \frac{1}{8}\left(\frac{\pi}{8}\right) + \frac{1}{32} = \boxed{\frac{\pi}{64} + \frac{1}{32}}$$

Numerical Approximation

Problem 1 (c)

(c) If Simpson's rule on four subintervals is used to approximate

$$\pi = \int_0^2 \frac{8}{x^2+4}\,dx,$$

a rational approximate value of π can be found as

$$\pi \approx \frac{1}{3}\left[1 + \frac{64}{a} + \frac{8}{b} + \frac{64}{c} + \frac{1}{d}\right]$$

where a, b, c, d are positive integers. Find a, b, c, d. [6 pts]

Solution to Problem 1 (c)

(c) Simpson's Rule Approximation: Approximate $\int_0^2 \frac{8}{x^2+4}\,dx$ with $n = 4$ subintervals. Step size $\Delta x = \frac{2-0}{4} = \frac{1}{2}$. Points: $0, \frac{1}{2}, 1, \frac{3}{2}, 2$. Let $f(x) = \frac{8}{x^2+4}$.

- $f(0) = 2$
- $f(\frac{1}{2}) = \frac{8}{1/4+4} = \frac{32}{17}$
- $f(1) = \frac{8}{5}$
- $f(\frac{3}{2}) = \frac{8}{9/4+4} = \frac{32}{25}$
- $f(2) = 1$

$$S_4 = \frac{\Delta x}{3}\left[f(0) + 4f\left(\frac{1}{2}\right) + 2f(1) + 4f\left(\frac{3}{2}\right) + f(2)\right]$$

Matching this with $\frac{1}{3}\left[1 + \frac{64}{a} + \frac{8}{b} + \frac{64}{c} + \frac{1}{d}\right]$, since $1 + \frac{1}{2} = \frac{3}{2}$, the $\frac{1}{d}$ term must be $\frac{1}{2}$. Thus, $\boxed{a = 17, \quad b = 5, \quad c = 25, \quad d = 2}$.

Reduction Formulas (2)

Problem 13 (a) & (b)

[20 pts] Let $I_n = \displaystyle\int_3^8 \frac{1}{x^n\sqrt{x+1}}\,dx$, where $n \geq 0$ is an integer.

(a) Evaluate I_0. [3 pts]

(b) Evaluate I_1. [6 pts]

Solution to Problem 13 (a) & (b)

Let $I_n = \int_3^8 \frac{1}{x^n\sqrt{x+1}}\,dx$.

(a) Evaluate I_0:

$$\begin{aligned} I_0 &= \int_3^8 \frac{1}{\sqrt{x+1}}\,dx = \left[2\sqrt{x+1}\right]_3^8 \\ &= 2(\sqrt{9}-\sqrt{4}) = 2(3-2) = \boxed{2} \end{aligned}$$

(b) Evaluate I_1: Let $u = \sqrt{x+1} \implies dx = 2u\,du$.
Limits: $x = 3 \to u = 2$; $x = 8 \to u = 3$. Also $x = u^2 - 1$.

$$\begin{aligned} I_1 &= \int_3^8 \frac{1}{x\sqrt{x+1}}\,dx = \int_2^3 \frac{1}{(u^2-1)u} 2u\,du = \int_2^3 \frac{2}{u^2-1}\,du \\ &= \int_2^3 \left(\frac{1}{u-1} - \frac{1}{u+1}\right)du = \Big[\ln|u-1| - \ln|u+1|\Big]_2^3 \\ &= (\ln 2 - \ln 4) - (\ln 1 - \ln 3) = \boxed{\ln\left(\frac{3}{2}\right)} \end{aligned}$$

Reduction Formulas (2)

Problem 13 (c) & (d)

(c) Apply integration by parts to show that [8 pts]

$$I_{n+1} = \frac{1-2n}{2n} I_n - \frac{3}{n8^n} + \frac{2}{n3^n}.$$

(d) Evaluate I_2. [3 pts]

Solution to Problem 13 (c) & (d)

(c) Apply Integration by Parts to I_n:
Let $u = x^{-n} \implies du = -nx^{-n-1}\,dx$.
Let $dv = (x+1)^{-1/2}\,dx \implies v = 2\sqrt{x+1}$.

$$I_n = \left[\frac{2\sqrt{x+1}}{x^n}\right]_3^8 - \int_3^8 2\sqrt{x+1}(-nx^{-n-1})\,dx$$

$$I_n = \left(\frac{6}{8^n} - \frac{4}{3^n}\right) + 2n\int_3^8 \frac{x+1}{x^{n+1}\sqrt{x+1}}\,dx$$

$$I_n = \frac{6}{8^n} - \frac{4}{3^n} + 2n\int_3^8 \frac{1}{x^n\sqrt{x+1}}\,dx + 2n\int_3^8 \frac{1}{x^{n+1}\sqrt{x+1}}\,dx$$

$$I_n = \frac{6}{8^n} - \frac{4}{3^n} + 2nI_n + 2nI_{n+1}$$

Solving for I_{n+1}:

$$2nI_{n+1} = (1-2n)I_n - \frac{6}{8^n} + \frac{4}{3^n} \implies \boxed{I_{n+1} = \frac{1-2n}{2n}I_n - \frac{3}{n8^n} + \frac{2}{n3^n}}$$

Comprehensive Problem (Part 1)

Part III

(a) & (b)

([25 points]) Let $I_n = \int_0^{\frac{\pi}{3}} \tan^n \theta \sec^3 \theta \, d\theta$, where $n = 0, 1, 2, \ldots$ is an integer.

(a) By writing $\tan^n \theta \sec^3 \theta = \tan^{n-1} \theta \cdot (\tan \theta \sec^3 \theta)$ and using integration by parts, show that for $n \geq 2$,

$$I_n = \frac{8(\sqrt{3})^{n-1}}{n+2} - \frac{n-1}{n+2} I_{n-2}$$

[10 pts]

(b) Evaluate I_0. [4 pts]

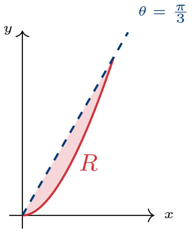
(Hint: You may need the integral of $\sec \theta$.)

Comprehensive Problem (Part 2)

(c) & (d)

(c) The region R in the first quadrant is enclosed by the polar curve $r = \sqrt{2\tan^2\theta\sec^3\theta}$, the ray $\theta = \frac{\pi}{3}$, and the polar axis ($\theta = 0$). Evaluate the exact area of R. [6 pts]

(d) If Simpson's rule on two subintervals is used to approximate I_0, the approximate value can be written as $I_0 \approx \frac{\pi}{a}\left[b + \frac{c\sqrt{3}}{9}\right]$. Find $a, b, c \in \mathbb{Z}^+$. [5 pts]



Solution to (a) - Part 1

Show: $I_n = \dfrac{8(\sqrt{3})^{n-1}}{n+2} - \dfrac{n-1}{n+2} I_{n-2}$

Let $u = \tan^{n-1}\theta \implies du = (n-1)\tan^{n-2}\theta \sec^2\theta \, d\theta$.
Let $dv = \tan\theta \sec^3\theta \, d\theta \implies v = \frac{1}{3}\sec^3\theta$.
Apply Integration by Parts ($\int u\,dv = uv - \int v\,du$):

$$I_n = \left[\frac{1}{3}\tan^{n-1}\theta \sec^3\theta\right]_0^{\frac{\pi}{3}} - \int_0^{\frac{\pi}{3}} \frac{1}{3}\sec^3\theta \cdot (n-1)\tan^{n-2}\theta \sec^2\theta \, d\theta$$

1. Evaluate the boundary term: At $\theta = \frac{\pi}{3}$, $\tan(\frac{\pi}{3}) = \sqrt{3}$ and $\sec(\frac{\pi}{3}) = 2$.
At $\theta = 0$, $\tan(0) = 0$ (since $n \geq 2$).

$$\text{Boundary} = \frac{1}{3}(\sqrt{3})^{n-1}(2)^3 - 0 = \frac{8}{3}(\sqrt{3})^{n-1}$$

Solution to (a) - Part 2

2. Simplify the integral term: Replace $\sec^2\theta$ with $1+\tan^2\theta$:

$$I_n = \frac{8}{3}(\sqrt{3})^{n-1} - \frac{n-1}{3}\int_0^{\frac{\pi}{3}} \tan^{n-2}\theta \sec^3\theta (1+\tan^2\theta)\, d\theta$$
$$I_n = \frac{8}{3}(\sqrt{3})^{n-1} - \frac{n-1}{3}\Big(I_{n-2} + I_n\Big)$$

Multiply the entire equation by 3 to clear denominators:

$$\begin{aligned} 3I_n &= 8(\sqrt{3})^{n-1} - (n-1)I_{n-2} - (n-1)I_n \\ (3+n-1)I_n &= 8(\sqrt{3})^{n-1} - (n-1)I_{n-2} \\ (n+2)I_n &= 8(\sqrt{3})^{n-1} - (n-1)I_{n-2} \end{aligned}$$

Divide by $(n+2)$ to obtain the final formula:

$$\boxed{I_n = \frac{8(\sqrt{3})^{n-1}}{n+2} - \frac{n-1}{n+2} I_{n-2}}$$

Solution to (b) & (c)

(b) Evaluate I_0:

$$I_0 = \int_0^{\frac{\pi}{3}} \sec^3\theta \, d\theta = \frac{1}{2}\Big[\sec\theta\tan\theta + \ln|\sec\theta + \tan\theta|\Big]_0^{\frac{\pi}{3}}$$

$$= \frac{1}{2}\left(2\sqrt{3} + \ln(2+\sqrt{3})\right) - 0 = \boxed{\sqrt{3} + \frac{1}{2}\ln(2+\sqrt{3})}$$

(c) Evaluate Polar Area:

$$A = \frac{1}{2}\int_0^{\frac{\pi}{3}} r^2 \, d\theta = \frac{1}{2}\int_0^{\frac{\pi}{3}} 2\tan^2\theta\sec^3\theta \, d\theta = \int_0^{\frac{\pi}{3}} \tan^2\theta\sec^3\theta \, d\theta$$

This is exactly I_2. Apply the reduction formula from (a) with $n = 2$:

$$I_2 = \frac{8(\sqrt{3})^1}{4} - \frac{1}{4}I_0 = 2\sqrt{3} - \frac{1}{4}\left(\sqrt{3} + \frac{1}{2}\ln(2+\sqrt{3})\right)$$

$$= \boxed{\frac{7\sqrt{3}}{4} - \frac{1}{8}\ln(2+\sqrt{3})}$$

Solution to (d)

(d) Simpson's Rule for I_0: Approximate $\int_0^{\frac{\pi}{3}} \sec^3\theta \, d\theta$ with 2 subintervals ($n = 2$).
Step size: $\Delta x = \frac{\pi/3-0}{2} = \frac{\pi}{6}$.
Points: $x_0 = 0$, $x_1 = \frac{\pi}{6}$, $x_2 = \frac{\pi}{3}$.
Function values $f(\theta) = \sec^3\theta$:

- $f(0) = 1^3 = 1$
- $f(\frac{\pi}{6}) = (\frac{2}{\sqrt{3}})^3 = \frac{8}{3\sqrt{3}} = \frac{8\sqrt{3}}{9}$
- $f(\frac{\pi}{3}) = 2^3 = 8$

Simpson's Formula: $S_2 = \frac{\Delta x}{3}[f(x_0) + 4f(x_1) + f(x_2)]$

$$S_2 = \frac{\pi/6}{3}\left[1 + 4\left(\frac{8\sqrt{3}}{9}\right) + 8\right] = \frac{\pi}{18}\left[9 + \frac{32\sqrt{3}}{9}\right]$$

Matching this with $\frac{\pi}{a}\left[b + \frac{c\sqrt{3}}{9}\right]$, we get:

$$\boxed{a = 18, \quad b = 9, \quad c = 32}$$